

Cosmic muons through the ESA test stand

A rate estimate with and without 3 ft of concrete

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CAD-registered geometry: eae30ac8f

Result. The current geometry gives approximately **1.41 muons/hour without an overhead slab** and **1.20 muons/hour with 3 ft of concrete**. Both numbers require a single muon to intersect all 26 active layers. They are conditional predictions from a sea-level spectrum and a straight-line continuous-slowng-down model, before detector efficiency; they are not measured rates or a full Geant4 transport result.

Overhead shielding	Muons/hour	Muons/day	Mean interval	$K_{\min}$ at surface
None	1.41	33.8	42.6 min	0.104 GeV
3 ft concrete	1.20	28.8	50.1 min	0.470 GeV

Here $K_{\min}$ is the acceptance-weighted mean kinetic energy needed to emerge from the modelled detector with at least 10 MeV remaining. That small residual is a range-table reliability floor, not an electronics threshold. The concrete reduces this model's rate by approximately **15%**.

What counts as going through everything?

A trajectory must have nonzero path length in an active volume in *every layer* listed below. Within a segmented layer, hitting any one bar is sufficient. A track need not hit all bars in that layer.

Subsystem	Required layers	Active volumes	Geometry used
Top HCal	2	24	Two crossed layers, 12 bars each
Trigger counters	6	72	Three stations, two staggered layers each
Silicon tracker	4	4	Active daughters; inactive borders excluded
LYSO target	2	33	Two staggered bar arrays
HCal above ECal	4	32	Four crossed layers, 8 bars each
ECal silicon	4	4	Four hexagonal active silicon layers
HCal below ECal	4	48	Four crossed layers, 12 bars each
Total	26	217	Bars and sensors are distinct volumes

The STEP model supplies the placement framework; ESA **ldmx-reduced-v3** supplies the ECal, trigger, tracker and target internals. HCal dimensions follow the test-stand branch. The inferred HCal bar counts and detector registration still need hardware confirmation. The support renderings are not additional material in the GDML used here.

The geometric bottleneck

The trigger bars are only 30 mm wide in X , while the HCal spans metres. After requiring the six trigger and two LYSO layers, the tracker and HCal do not remove further tracks in this integration; the displaced ECal footprint reduces the remaining rate by about 36%. The accepted rays are close to vertical: 90% are within 4.82° and 99% within 6.28° under the reference angular model.

For the simple reference intensity $I(\theta) = 70 \cos^2 \theta \text{ m}^{-2}\text{s}^{-1}\text{sr}^{-1}$, associated with sea-level muons above 1 GeV/ c [1],

$$G_2 = \int dA d\Omega \cos^3 \theta \mathbf{1}_{\text{all}} = 4.0021 \times 10^{-6} \text{ m}^2\text{sr}, \quad R_{p>1}^{\text{geom}} = 1.0085 \pm 0.0004 \text{ hour}^{-1}.$$

The quoted error is numerical integration error only. The headline rates are larger because the energy-dependent calculation also includes lower-energy muons that have enough range to cross the selected materials.

Geometry and shielding illustrations

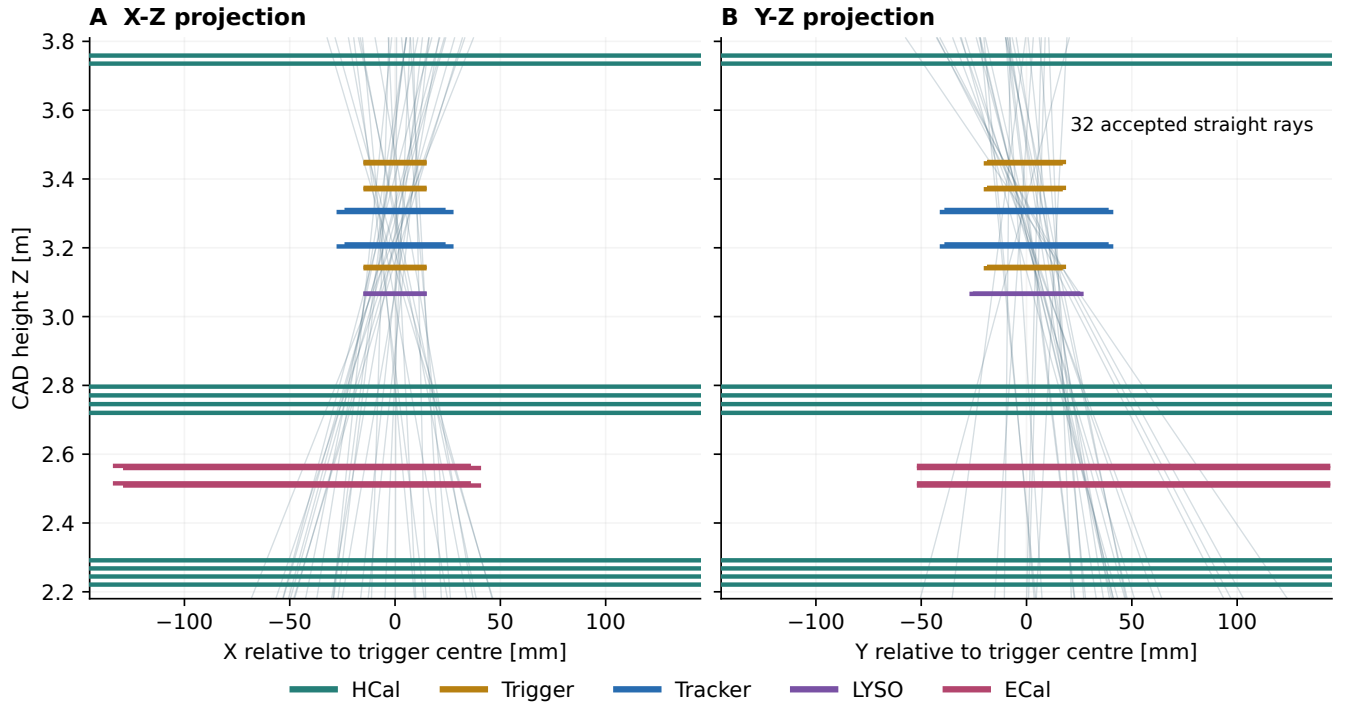


Figure 1. Two projections of 32 accepted sample trajectories. Coloured segments mark the actual layer centres and transverse extents; their line thickness is enlarged for legibility. HCal bars extend beyond the displayed transverse window. The ECal offset and narrow trigger/LYSO corridor restrict the accepted directions. These are straight geometric rays, not simulated scattering histories.

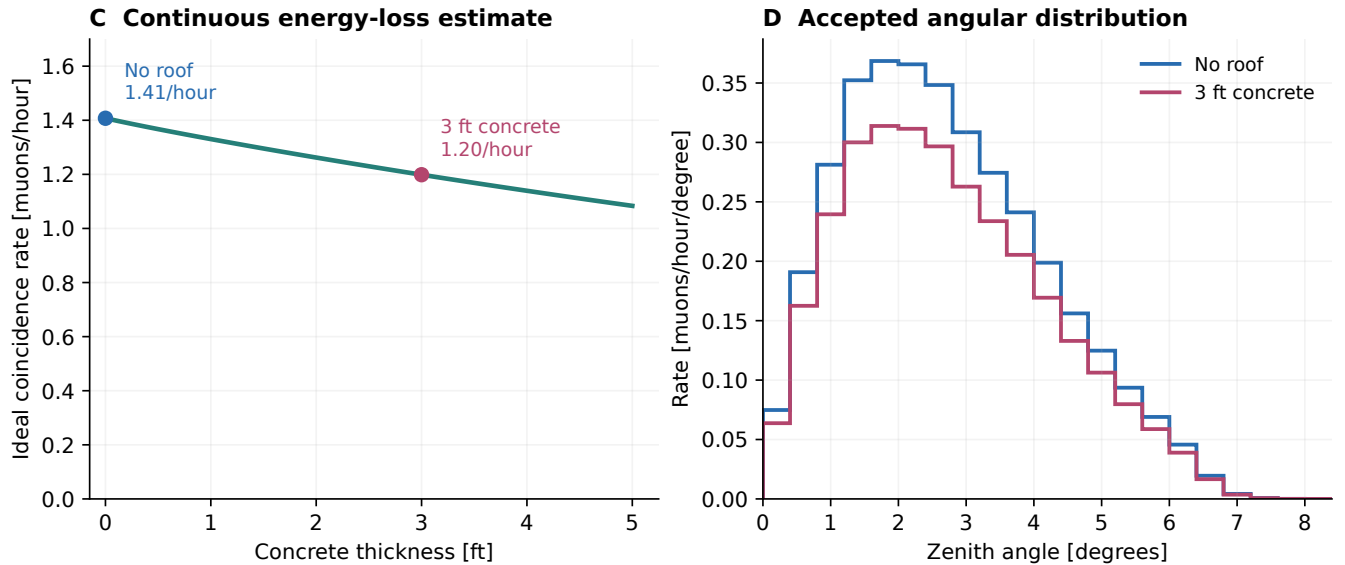


Figure 2. The continuous-loss estimate versus concrete thickness, and the angular distribution of muons that satisfy all 26 layer requirements. The 3 ft comparison assumes a uniform horizontal slab that covers the stand and all accepted directions, with concrete density 2.30 g cm^{-3} [3]. Roof height and openings are not modelled. No efficiency correction or calibrated uncertainty band is implied.

Calculation and checks

Geometry integration. Two points are sampled on horizontal planes through the upper trigger and lower LYSO layer, separated by $D = 383.4$ mm. Sampling rectangles are slightly expanded to include tracks that clip a finite-thickness anchor volume. If $\mathbf{u} = (\mathbf{b} - \mathbf{a})/D$, then

$$R = \int d^2a d^2b \frac{I(\theta) \cos^4 \theta}{D^2} \mathbf{1}_{\text{all}}, \quad \cos \theta = (1 + |\mathbf{u}|^2)^{-1/2}.$$

This includes both the projected-area factor and the solid-angle Jacobian. For $I = I_0 \cos^2 \theta$, each ray carries a $\cos^6 \theta$ weight. Box and hexagonal-prism intersections retain the physical bar gaps, stagger, tracker rotations and finite layer thickness. Eight independently scrambled Sobol sequences of 2^{18} point pairs give 2,097,152 trials. An independent pseudorandom estimate agrees within its numerical error. ROOT's original GDML solids agree with all 4,992 individual layer decisions in a separate 192-ray test.

Spectrum. The energy-dependent model uses Guan et al.'s modified Gaisser spectrum[2], with total energy E in GeV and j in $\text{cm}^{-2}\text{s}^{-1}\text{sr}^{-1}\text{GeV}^{-1}$:

$$j(E, \theta) = C 0.14 \left[E + \frac{3.64}{(\cos \theta^*)^{1.29}} \right]^{-2.7} \left[\frac{1}{1 + 1.1E \cos \theta^*/115} + \frac{0.054}{1 + 1.1E \cos \theta^*/850} \right].$$

The factor $C = 1.1595$ sets the vertical integral above $p = 1$ GeV/ c to $70 \text{ m}^{-2}\text{s}^{-1}\text{sr}^{-1}$. The published curvature correction for θ^* is implemented in the supplied script. The flux above each ray's required surface energy replaces the reference $70 \cos^2 \theta$ weight. The resulting lower-energy contribution is model dependent.

Stopping model. The supplied 3 ft thickness is 91.44 cm, or $210.312 \text{ g cm}^{-2} = 2.10312 \text{ m}$ water equivalent at the assumed density. ROOT chords and the GDML densities give a mean vertical detector column of 32.70 g cm^{-2} over 96 accepted rays, of which roughly 20.6 g cm^{-2} is HCal scintillator. For this estimate, the whole detector column is represented by a polystyrene CSDA range[4]; the roof uses the concrete CSDA table[3]. This is an approximation to the mixed detector materials, not material-by-material transport. With tabulated range functions $\mathcal{R}_p$ and $\mathcal{R}_c$,

$$K_d = \mathcal{R}_p^{-1}[\mathcal{R}_p(10 \text{ MeV}) + X_d \sec \theta], \quad K_s = \mathcal{R}_c^{-1}[\mathcal{R}_c(K_d) + X_c \sec \theta].$$

The flux integral starts at $E = K_s + m_\mu$. With no roof, $K_s = K_d$. Doubling the detector column changes the 3 ft prediction from 1.20 to 1.17/hour; requiring 100 MeV at the final exit gives 1.17/hour.

Interpretation. The recorded coincidence rate will also depend on the light-yield thresholds, dead channels, trigger logic and live time. Multiple scattering, decay and stochastic energy losses are omitted. Grazing tracks can count here: requiring a hit at every layer's midplane instead reduces the reference rate from 1.0085 to 0.853/hour, showing that the signal threshold matters. The flux normalization itself is not exact; the PDG review notes measurements 10–15% below the reference value[1]. No combined systematic uncertainty or local flux calibration is claimed. The estimate is for cosmic muons of both charges, not all cosmic-ray species or shower coincidences.

Reproduction. The folder contains the geometry snapshot, scripts, numerical tables, range data, figures and this L^AT_EX source. Run `build_report.sh` with Python (NumPy, SciPy, Matplotlib), ROOT and Tectonic available. The SHA256SUMS file records artifact identities. The original STEP SHA-256 and all source GDML hashes are preserved in `geometry/provenance.json`.

References

- [1] Particle Data Group, *Cosmic Rays*, 2022 review, Sec. 30.3.1. <https://pdg.lbl.gov/2022/reviews/rpp2022-rev-cosmic-rays.pdf>.
- [2] M. Guan et al., *A parametrization of the cosmic-ray muon flux at sea-level*, arXiv:1509.06176 (2015), Eqs. (2)–(3). <https://arxiv.org/abs/1509.06176>.
- [3] PDG, *Shielding concrete: properties and muon CSDA range*. https://pdg.lbl.gov/2022/AtomicNuclearProperties/HTML/shielding_concrete.html.
- [4] PDG, *Muon stopping power and range in polystyrene*. https://pdg.lbl.gov/2019/AtomicNuclearProperties/MUE/muE_polystyrene.txt.